METHOD STATEMENT FOR
AIR SUPPLY & IMPULSE LINE TESTING

(Doc No. CT8-PRO-CM-018)

USA PROJECT UNIT 1 - TALLGRASS

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1. SCOPE

The purpose of this method statement is to describe the steps of the activities and methods to executing instrumentation scope of Air supply and Impulse installation and leaks tested the project. This document to conjunction with the inspection methods for instrumentation and electrical in accordance with all applicable drawings, standards, codes and related specification requirement in the project scope.

2. REFERENCE DOCUMENTS

The Instrumentation for installation refers to the Codes, Standards and Regulations shall be followed as listed.

National Fire Prevention Association (NFPA)

- NFPA 70 - National Electrical Code (NEC)
- NFPA 70E - Standard for Electrical Safety in Workplace
- NFPA 101 - Life Safety Code
- NFPA 497 - Recommended Practice for the Classification of Flammable Liquids, Gases, or Vapors and of Hazardous (Classified) Locations for Electrical Installation in Chemical Process Area

Occupational Safety and Health Administration (OSHA)

- OSHA 1926.56

Underwriters' Laboratories (UL)

- UL50

The Instrumentation for cable installation refer to the Engineering Desing Standard and MDLs drawing shall be followed as listed.

- MDL-201 Modular arrangement of Instrumentation
- MDL-202 Instrument Tubing routing and support detail drawings
- MDL-203 Instrument Hook-up Drawings
- MDL-204 Instrumentation Name Plates Drawing
- MDL-205 Instrument List for all installed equipment
- MDL-207 Arrangement of Stanchion and Tubing around GT Vent Fans
- MDL-208 Material Requisition for Instrument Tubing and Fitting
- MDL-209 Material Requestion for Instrument Stanchion
- MDL-211 Instrument Overall Bom
- MDL-301 Engineering Desing Standard for Electrical installation

3. RESPONSIBILITIES

Project Manager (PM)

Project Manager has overall responsibility for achieving the progress & quality of the Instrumentation installation works under the safety requirements through close monitoring on the section managers' and subcontractors' performances.

Construction Manager (CM)

The construction manager is responsible for overall construction activities including, but not limited to, safe operations, scheduling activities, coordinating the work, material requests, quality, productivity, work front, manpower and equipment allocation.

Health, Safety & Environment Supervisor

Ensure that clearance certificate and work permit are available to commencement of any work. Check personnel's compliance to safety requirement, such as the wearing of Personal Protection Equipment (PPE), Tool box talk (TBM). Check work for any unsafe conditions. Liaise with the field and Environment (HSE) Plan/Procedure.

Section Manager or Superintendent

Section managers or superintendent shall effectively manage the staff daily activities in order to achieve the progress & quality of the instrument tubing assembly and installation works under the safety requirements, and he also shall ensure that following requirements shall be met;

- Ensure manpower's works shall meet the quality requirements.
- Ensure all required inspections and tests are witnessed by QC/Owner.
- Perform JSA, provide the safety measures and monitor the working safety.
- Assign the qualified personnel if required by quality or safety procedures.

QC Inspector

In conjunction with section supervisor, he shall be responsible for the quality of Instrumentation equipment assembly and installation works that includes material verifications, personnel qualifications, in-process inspections, punch inspections, coordination for owner's witness, etc.

4. WORK PROCEDURE

4.1 GENERAL PIPING AND TUBING INSTALLATION

Instrument piping (or tubing) should be installed between the instrument and detecting root valve or the instrument and root valve from air header (common to impulse piping and instrument air piping).

Instrument air and instrument sensing lines shall be routed to prevent the entrapment of liquids; all lines shall have low point drains.

Tubing supports shall be provided with rubber insert.

Stainless steel tube joint shall be compression type fitting. However, instrument manifold and instrument are excluded. At least 50mm shall be obtained for the straight tube portion at the terminal section.

Welding of steel fitting shall be by TIG welding.

The bend part of the impulse pipe (or tube) shall be of the cooling tube bending fabrication

Pipe jointing shall be conducted by Welding. Tube jointing shall be conducted by compression tube fittings.

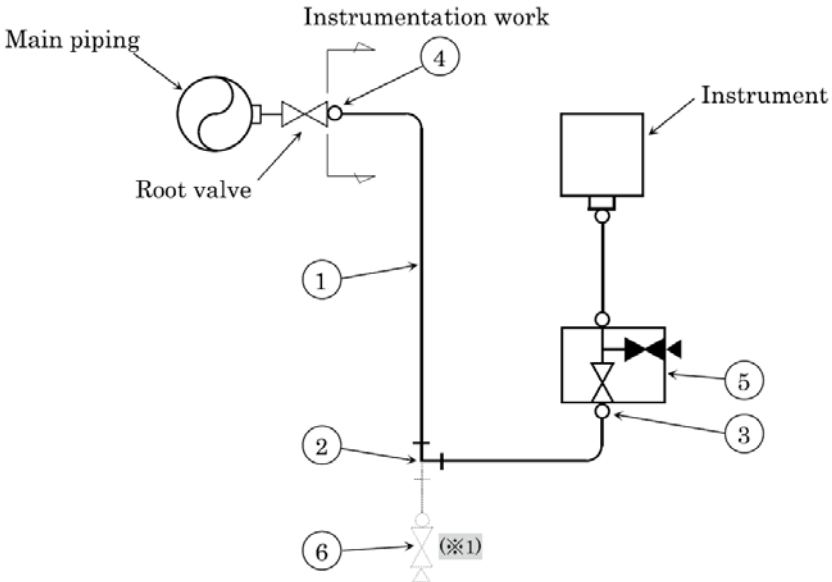
Approved and suitable thread sealant to be used as required (for NPT threads)

Supports of piping (or tubing) shall be “39.4~78.'7” (lm ~ 2m) intervals.

4.2 DETAIL PIPING AND TUBING INSTALLATION

4.2.1 Arrangement of impulse Piping and Tubing

- Design criteria of the impulse piping and tubing



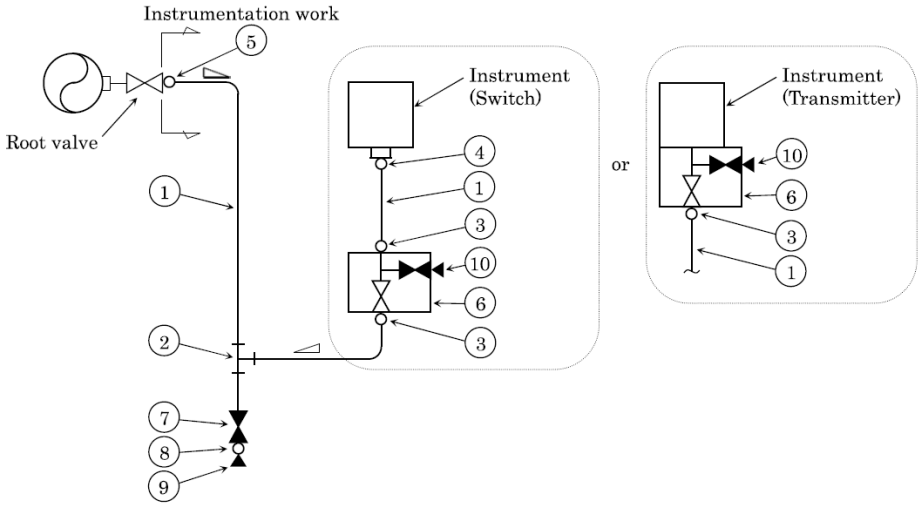
(※1) : Blow valve should be provided for the water and air system if necessary.

No.	Section	Design Pressure	Design Temperature
1	Impulse piping (or tubing)	Main piping press.	Process fluid temp.
2	Fitting (or tube fitting)	Main piping press.	Process fluid temp.
3	Connector (or tube connector)	Main piping press.	Process fluid temp.
4	Weld connector	Main piping press.	Process fluid temp.
5	Instrument valve	Main piping press.	Saturated temp. at main line press.
6	Blow valve	Main piping press.	Process fluid temp.

Note: As for the details of the impulse piping and tubing refer to following drawings which will be submitted separately DWG. No. MDL-203 Instrument Hook-up Drawings.

Typical drawing of impulse piping and tubing for water system is shown in Fig.4-16 and 17.

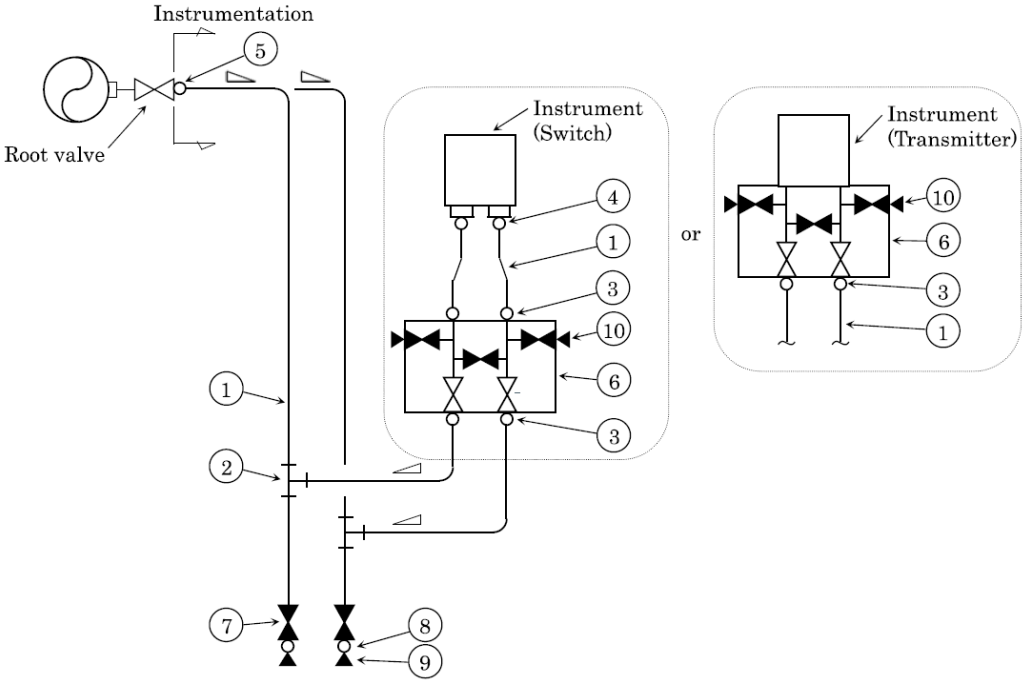
For pressure instrument except for pressure gauge



1) Tube 1/2” OD x 0.065”t
2) Union tee 1/2” OD
3) Male connector 1/2” OD x NPT1/2(M)
4) Male connector or Female connector 1/2” OD x suitable size for instrument
5) Weld connector suitable size for root valve x 1/2” OD

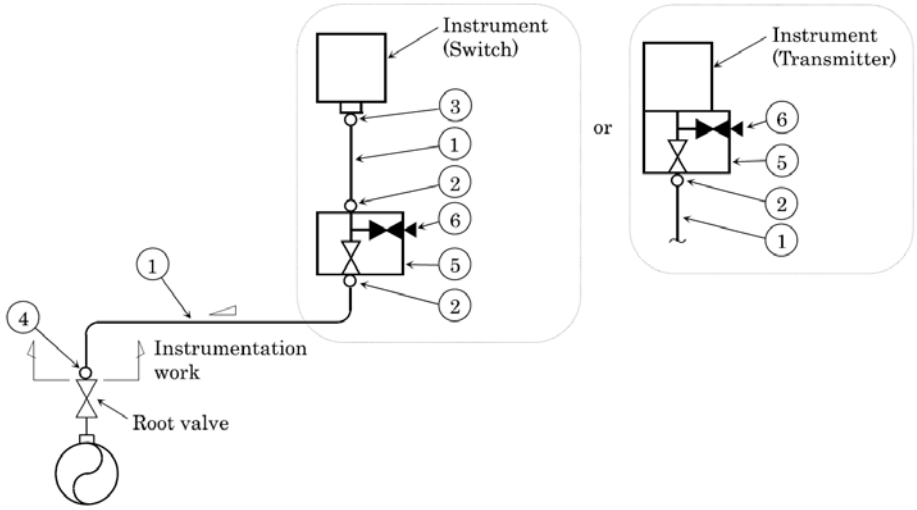
6) 2-Valve manifold (NPT1/2(F)(Process) x NPT1/2(F) or NPT1/4(F)(Vent)
7) Blow down valve 1/2" OD
8) Female adapter 1/2" OD x NPT1/2(F)
9) Plug NPT1/2(M)
10) Plug NPT1/4(M)

Typical drawing for differential pressure instrument



1) Tube 1/2" OD x 0.065"t
2) Union tee 1/2" OD
3) Male connector 1/2" OD x NPT1/2(M
4) Male connector or Female connector (1/2" OD x suitable size for instrument
5) Weld connector suitable size for root valve x 1/2" OD
6) 5-Valve manifold NPT1/2(F)(Process) x NPT1/2(F) or NPT1/4(F)(Vent)
7) Blow down valve 1/2" OD
8) Female adapter 1/2" OD x NPT1/2(F)
9) Plug (NPT1/2(M)
10) Plug (NPT1/4(M)

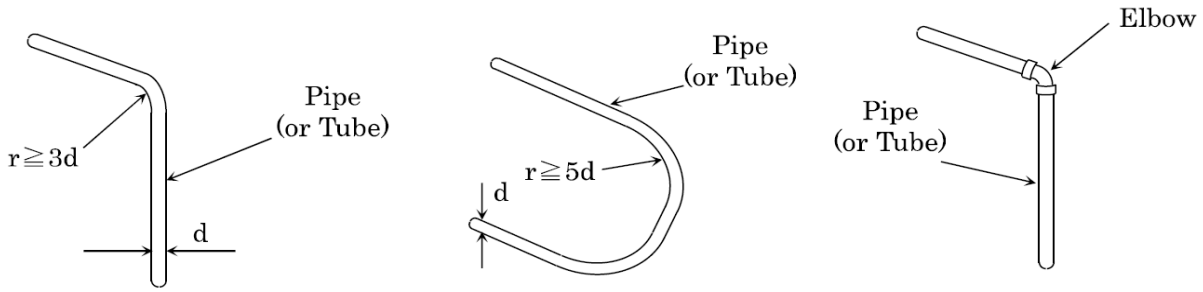
Typical drawing of impulse piping and tubing for air system.



1) Tube 1/2”OD x 0.065”t
2) Male connector 1/2”OD x NPT1/2(M)
3) Male connector or Female connector 1/2”OD x suitable size for instrument
4) Weld connector suitable size for root valve x 1/2”OD
5) 2-Valve manifold NPT1/2(F)(Process) x NPT1/2(F) or NPT1/4(F)(Vent),
6) Plug (NPT1/4(M), A182F316 or better

4.2.2 Bending radius

The radius of bending should be greater than or equal to 5 times the outside diameter of pipe (or tube) as shown in Fig.4-1 (i), (ii). When the radius of bending becomes too small, an elbow fitting should be used as shown in Fig.4-1(iii).



(i) 90°or less bending
r : Radius of bending
d : Outside diameter

(ii) U bending

(iii) Use of elbow fitting

Fig.4-1 : Bending radius of piping and tubing

4.2.3 Jointing for piping work

Screw threads should be equivalent to taper pipe thread.

Threading should be conducted by threading machine, when threading is done by die stock, a lubricant should be used, after threading, grease and the like shall be wiped off with waste cloth.

Threaded portion should be totally sealed with sealing tape or pasty sealant.

4.2.4 Weld jointing

Welding of steel fitting should be by TIG welding.

Socket welding should be used.

Welding rods that meet specific welding purposes should be used.

Welding should be performed on straight portions.

Welding should be conducted after tack welding.

For socket welding, leg length should be greater than or equal to 1.25 times the pipe wall thickness and clearance after socket welding should be 0.08" ±0.04" (2 ±1mm) as shown in Fig.4-5.

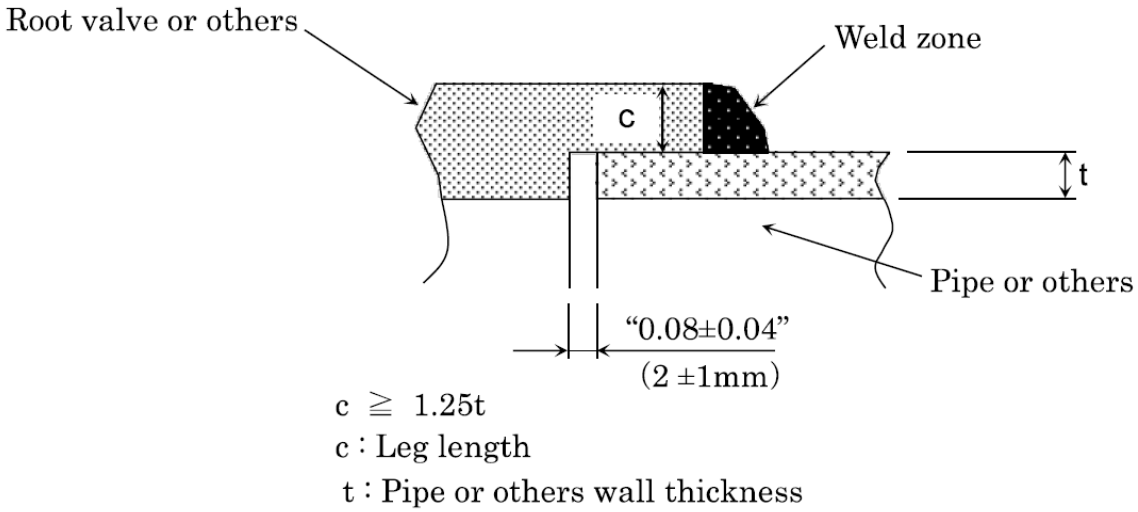
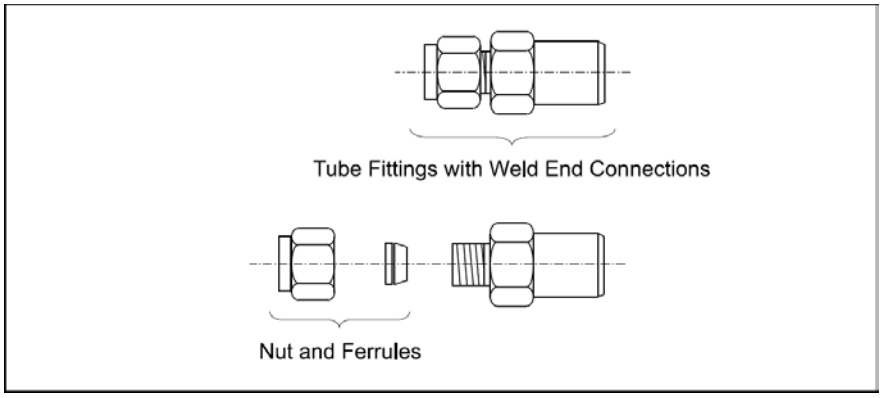


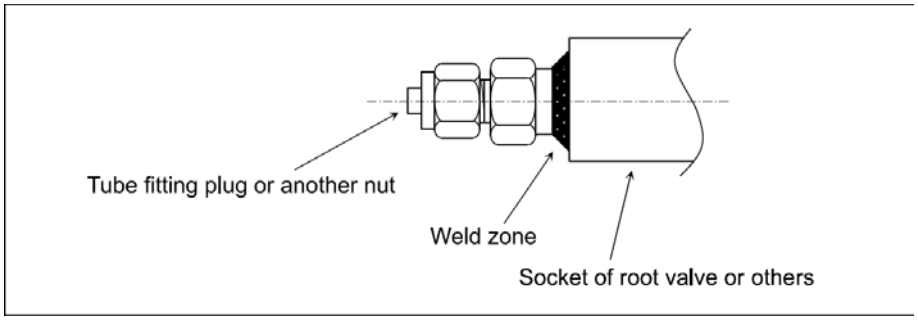
Fig.4-5 : Socket welding

Welding details for tube fitting with weld end connections should be used as shown in bellow,

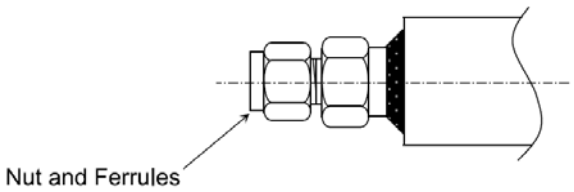
- Remove the nut and ferrules.



- Turn a tube fitting plug or another nut onto the fitting so that it is finger-tight, this protects the threads and sealing components.
- Provide a suitable heat sink to dissipate the heat.
- Tack weld at four positions 90°apart to hold the fitting in place and to ensure alignment and
- concentricity of the components.
- Complete the weld in accordance with welding procedure.



Remove the plug or nut and replace the nut and ferrules.



4.2.5 Jointing for tubing work

Stainless steel tube joint should be compression type fitting; However, instrument manifold and instrument are excluded, at least 50mm should be obtained for the straight tube portion at the terminal section as shown in Fig.4-6

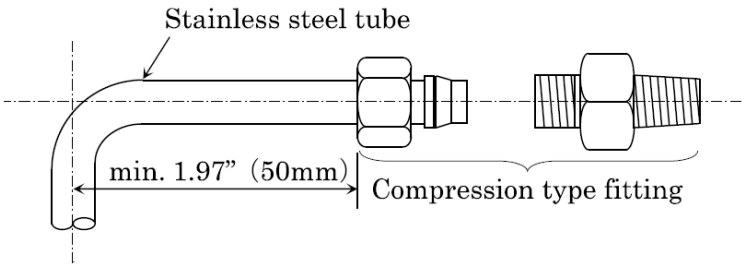


Fig.4-6 : Terminal section of tubing

The threaded portion of fitting should be tied around with teflon seal tape as shown in Fig.4-7. In this case, the threaded portion for a distance of "0.08～0.12" (2 ~ 3mm) from the end should be free from the seal tape.

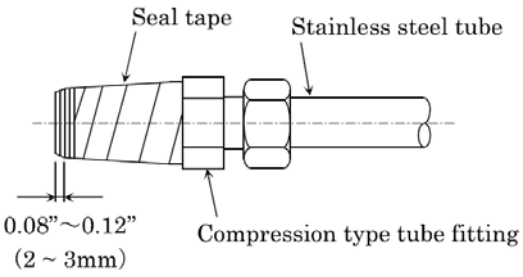


Fig.4-7 : Sealing of terminal fitting

Installation instructions for Hy-Lok and Swagelok tube fittings, these instructions apply both to traditional fitting and to fittings with the advanced back-ferrule geometry.

Fully insert the tube into the fitting and against the shoulder as shown in Fig.4-8.: rotate the nut finger-tight. High-pressure applications and high safety-factor systems: Further tighten the nut until the tube will not turn by hand or move axially in the fitting.

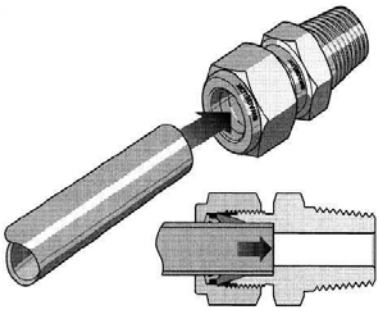


Fig.4-8 : Method of the insert the tube into the fitting and against the shoulder

Mark the nut at the 6 o'clock position as shown in Fig.4-9.

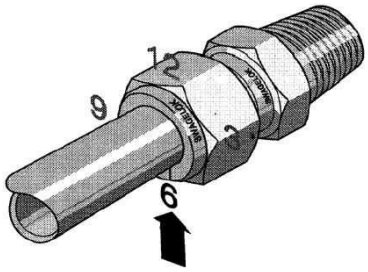


Fig.4-9 : Marking position

While holding the fitting body steady, tighten the nut one and quarter turns to the 9 o'clock position. For 1/16, 1/8, and 3/16 in; 2,3, and 4mm tube fittings, tighten the nut three-quarters turn to the 3 o'clock position as shown in Fig.4-10.

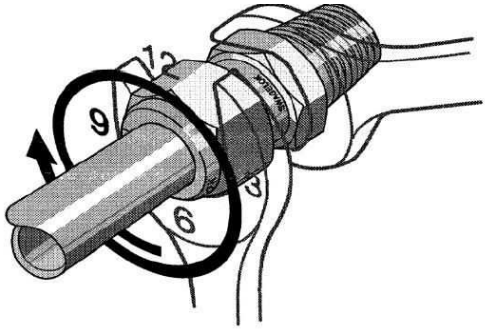
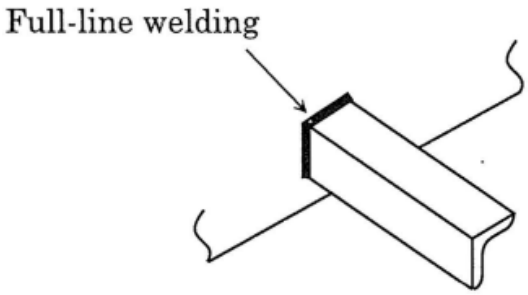


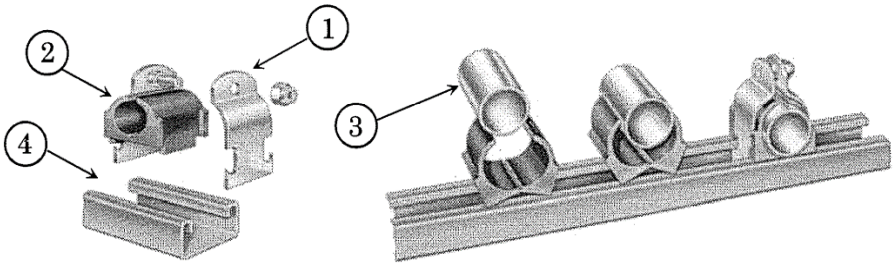
Fig.4-10 : Method of tighten the nut

4.2.6 Supporting for Pipe and tubing

- When supporting bracket is welded to structure, full line welding should be used.

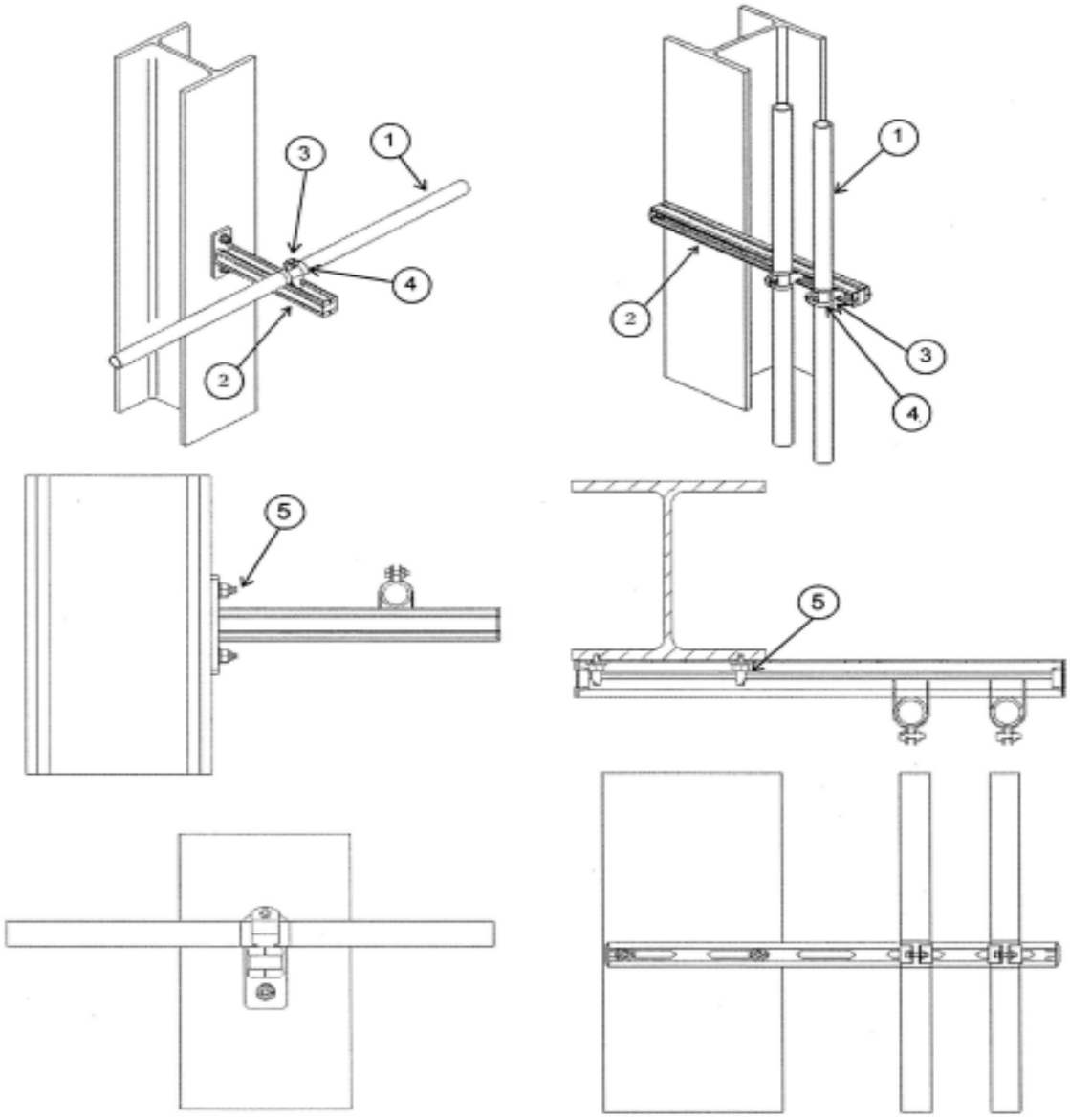


- Tube clamp should be installed for support



① Clamp (Stainless steel)
② Cushion (Thermoplastic polypropylene-based elastomer)
③ Tube Stainless steel)
④ Steel channel (Steel)

- In case of supporting bracket, structure clamp, HILTI S-BT or equivalent shall be used



No.	Name	Remarks
1	Piping(or tubing)	Stainless steel
2	Strut channel	Hot-dip galvanized or Stainless steel
3	Clamp	Hot-dip galvanized or Stainless steel
4	Anti corrrosion sheet	Stainless steel
5	HILTI S-BT(Stud bolt)	Hot-dip galvanized or Stainless steel

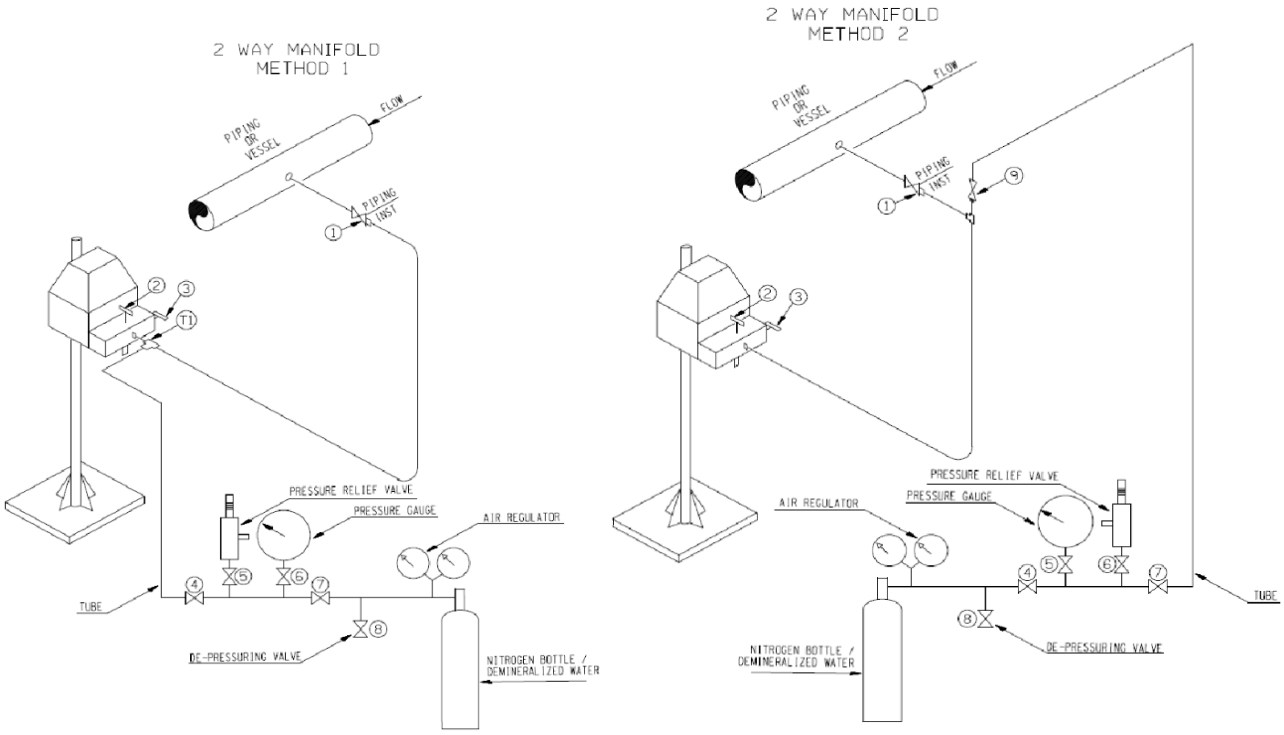
4.3 INSTRUMENTATION PIPING AND TUBING LEAK TEST

The Instrument impulse lines, whether tube or pipe, between the process main block valve and instrument isolation manifold should be tested using the nitrogen gas at times the pressure of the piping system or process equipment to which it connected.

Any instrument that could be subjected to damage to test shall be disconnected from the system. Instrumentation impulse line shall be tested by closed the process block valve and safely assembled main necessary test equipment's. The leaks, which cause the test pressure to decrease more than one graduation on test gauge during the holding time, shall not be acceptable.

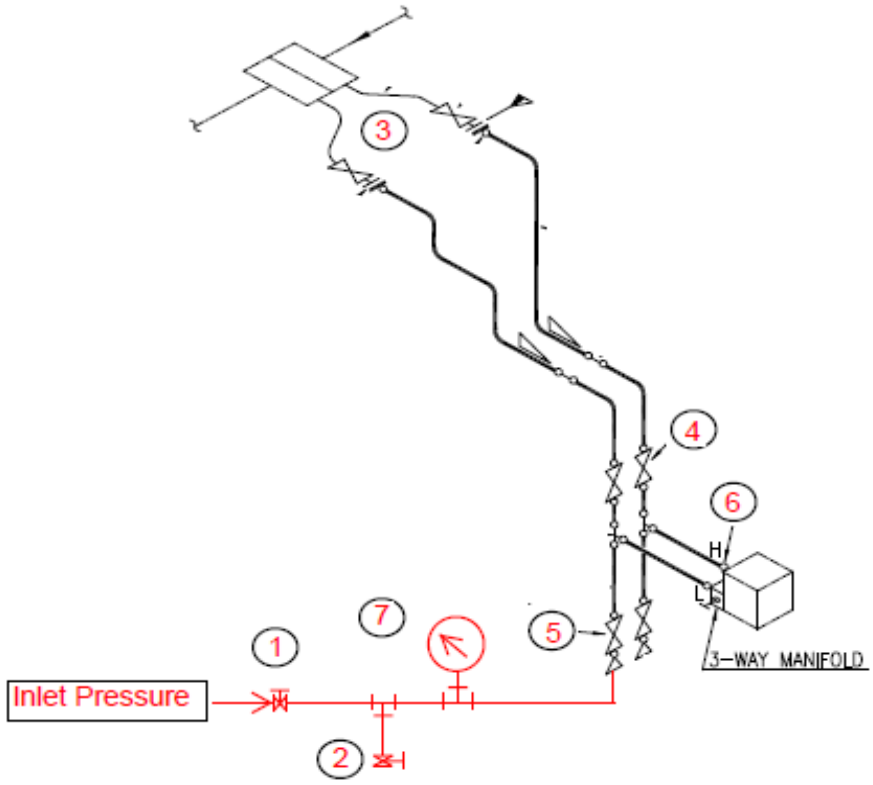
All impulse line should be cleaned by blowing dry air or nitrogen testing gas before start the leak test. Any observed leakage shall be corrected and re-tested of the system shall be carry out until getting a satisfactory result. After the test, have been completed, the instrumentation shall be identified with a color code sticker.

- Diagram Leak Test Impulse for 2 Manifold Method 1 And 2



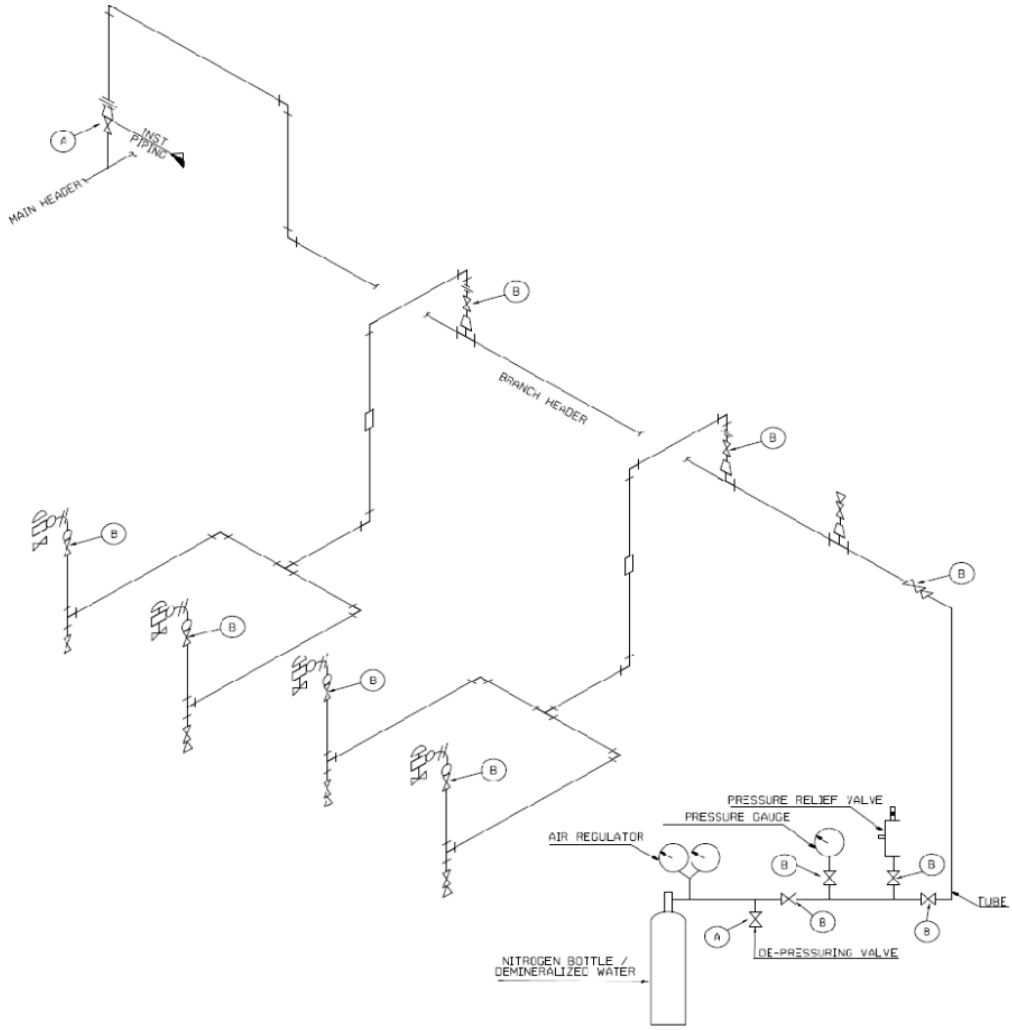
Close Valves 1, 2, 3 and 8. Open valves 4, 5, 6 and 7 then pressurized the system up to required test pressure (in-case method 1) Close valve 1, 2, 3 and 8. Open 4, 5, 6, 7 and 9 then pressurized the system up to required test pressure (in case method 2).

- Diagram Leak Test for 3-WAY Manifold valve Method.



After installation is complete, equipped for Leak Test (Diagram), Off Valve (1), (2), (3) and Manifold Valve (6) to properly, Open Valve (4) and (5), and gradually open Valve (1) to increase the pressure by reading the voltage at Pressure Gauge (7) until the voltage as needed Leak Test checked for leaks by the Tube connect by using reagents Check Leak.

- Diagram Leak Test for Airline piping



Close Valves “A”. Open valves B and disconnect the tube at valve regulator I/P side and install OD plug on the tube then pressurized the system up to required test pressured.

Bubble leak test shall be carried out for Instrument air tube fittings of I/P regulator side with permanent instrument air at site.

Confirm there is no leak by pouring soap water or snoop leak solution on each connection.

Instrument impulse lines shall be leak tested for at least 10 minutes duration time.

All instrument process tubing through all drain valves shall be cleaned by air blowing before and/or after the test. De-pressurize the system after the test by releasing the pressure in the De- pressuring valve.

All tests shall be witnessed by Client Company Representative. After successful completion of each testing, the airline piping shall be marked with a yellow sticker.

QC inspection and test results shall be filled and recorded in applicable forms

Confirm there is no leak by pouring soap water or snoop leak solution on each connection.

All instrument airline piping through all drain valves shall be cleaned by air blowing before the test.

All joints which are found defective shall be repaired immediately and be re-tested by the original test method.

QC inspection and test results shall be filled and recorded in applicable forms.

1 For pneumatic test at ***1.25 times*** of the design pressure / maximum operating pressure.

5. QUALITY CONTROL

Electrical and Instrument Installation shall be performed in accordance with the specification and Drawing. The inspection, details the acceptance criteria for each equipment installation, lists the relevant specifications, inspection procedures, test frequency, inspection characteristics and the subsequent verifying records. The Electrical and Instrument inspector shall prepare and submit on a daily basis whilst carrying out works pertained in this document checklists indicating the completed significant activities, location of their works and confirmation of the quality checks performed to refer **APPENDIX-A**

5.1 APPENDIX-A (INSPECTION PERORT)